

PSET 7: General random variables

Problem 1: Spot the pdf/CDF

- a. $f(x) = 3x^2$ on $(0, 1)$.

Non-negativity: $3x^2 \geq 0$ for all $x \in (0, 1)$. ✓

Normalization:

$$\int_0^1 3x^2 dx = \left[x^3 \right]_0^1 = 1^3 - 0^3 = 1.$$

Both pdf conditions hold, so this is a **valid pdf**.

- b. $F(x) = x/6$ on $[0, 6]$ (0 before, 1 after).

Nondecreasing: $F'(x) = 1/6 > 0$ on $(0, 6)$, and F is constant (hence non-decreasing) outside. ✓

Limits: $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. ✓

Continuity at the junctions: at $x = 0$, $x/6 \rightarrow 0 = F(0)$; at $x = 6$, $x/6 \rightarrow 1 = F(6)$. ✓

All CDF conditions hold, so this is a **valid CDF**.

- c. $f(x) = x/2$ on $(0, 3)$.

Check as a PDF: $f(x) = x/2 \geq 0$ on $(0, 3)$, so non-negativity holds. But

$$\int_0^3 \frac{x}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_0^3 = \frac{1}{2} \cdot \frac{9 - 0}{2} = \frac{9}{4} = 2.25 \neq 1.$$

The integral is not 1, so normalization fails: **not a valid pdf**.

Check as a CDF: a CDF must satisfy $0 \leq F(x) \leq 1$ for every x . But $f(2) = 1$ and $f(3) = 1.5 > 1$, so this function takes values above 1 on $(0, 3)$. That alone violates the upper-bound requirement, so it is also **not a valid CDF**.

Since it satisfies neither set of conditions, this function is **neither** a pdf nor a CDF.

- d. $F(x) = x^3$ on $[0, 1]$ (0 before, 1 after).

Nondecreasing: $F'(x) = 3x^2 \geq 0$ on $(0, 1)$. ✓

Limits: $F \rightarrow 0$ as $x \rightarrow -\infty$ (it is 0 for $x < 0$) and $F \rightarrow 1$ as $x \rightarrow \infty$ (it is 1 for $x \geq 1$). ✓

Continuity at the boundaries: at $x = 0$, $0^3 = 0 = F(0)$; at $x = 1$, $1^3 = 1 = F(1)$. ✓

All CDF conditions hold, so this is a **valid CDF**.

- e. $f(x) = 2 - x$ on $(0, 3)$.

Check as a PDF: at $x = 3$, $f(3) = 2 - 3 = -1 < 0$. Non-negativity fails immediately, so this is **not a valid pdf**. (Since it already fails, the normalization integral does not need to be checked.)

Check as a CDF: $f'(x) = -1 < 0$ on $(0, 3)$, so the function is strictly *decreasing*, not nondecreasing. This alone rules it out as a CDF. In addition, $f(0^+) = 2 > 1$, which violates the requirement $F(x) \leq 1$. Both violations independently rule this out as a **valid CDF**.

Since it satisfies neither set of conditions, this function is **neither** a pdf nor a CDF.

- f. For the two functions identified as CDFs in (b) and (d), obtain the corresponding pdf by differentiating:

From (b): $F(x) = x/6$ on $[0, 6]$, so

$$f(x) = F'(x) = \frac{d}{dx} \left(\frac{x}{6} \right) = \frac{1}{6}, \quad 0 < x < 6, \quad f(x) = 0 \text{ otherwise.}$$

From (d): $F(x) = x^3$ on $[0, 1]$, so

$$f(x) = F'(x) = \frac{d}{dx} (x^3) = 3x^2, \quad 0 < x < 1, \quad f(x) = 0 \text{ otherwise.}$$

- g. Functions (a) and (d) are a pdf/CDF pair for the same random variable, since differentiating the CDF in (d) gives

$$\frac{d}{dx} (x^3) = 3x^2,$$

which is exactly the pdf given in (a). Their supports also match: $(0, 1)$ in both cases.

Problem 2: Identifying the pdf

- a. Prices are uniform on $[160,000, 340,000]$. The width of this interval is

$$b - a = 340,000 - 160,000 = 180,000,$$

so

$$f(x) = \begin{cases} \frac{1}{180,000} & 160,000 \leq x \leq 340,000 \\ 0 & \text{otherwise} \end{cases}$$

- b. Using $E[X] = \frac{a+b}{2}$ and $Var(X) = \frac{(b-a)^2}{12}$ for $X \sim \text{Uniform}(a, b)$:

$$E[X] = \frac{160,000 + 340,000}{2} = \frac{500,000}{2} = 250,000$$

$$Var(X) = \frac{(180,000)^2}{12} = 2,700,000,000$$

$$\sigma = \sqrt{Var(X)} = \sqrt{2,700,000,000} \approx 51,961.524$$

Problem 3: Parliamentary elections

- a. Setting the integral of the pdf equal to 1:

$$\int_0^4 kx^4 dx = k \left[\frac{x^5}{5} \right]_0^4 = k \cdot \frac{1024}{5} = 1 \implies k = \frac{5}{1024}$$

- b. Integrating the pdf from 0 to x (for $0 \leq x < 4$):

$$F(x) = \int_0^x \frac{5}{1024} t^4 dt = \frac{5}{1024} \left[\frac{t^5}{5} \right]_0^x = \frac{5}{1024} \cdot \frac{x^5}{5} = \frac{x^5}{1024}$$

So

$$F(x) = \begin{cases} 0 & x < 0 \\ x^5/1024 & 0 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

- c. First moment:

$$E[X] = \int_0^4 x \cdot \frac{5}{1024} x^4 dx = \frac{5}{1024} \left[\frac{x^6}{6} \right]_0^4 = \frac{5}{1024} \cdot \frac{4096}{6} = \frac{10}{3} \approx 3.333$$

Second moment:

$$E[X^2] = \int_0^4 x^2 \cdot \frac{5}{1024} x^4 dx = \frac{5}{1024} \left[\frac{x^7}{7} \right]_0^4 = \frac{5}{1024} \cdot \frac{16384}{7} = \frac{80}{7} \approx 11.429$$

Variance:

$$Var(X) = E[X^2] - (E[X])^2 = \frac{80}{7} - \left(\frac{10}{3} \right)^2 = \frac{720}{63} - \frac{700}{63} = \frac{20}{63} \approx 0.317$$

- d. Setting $F(x) = 1/2$:

$$\frac{x^5}{1024} = \frac{1}{2} \implies x^5 = 512 \implies x \approx 3.482$$

- e. Since X is a continuous random variable, its pdf assigns zero probability to any single point:

$$\Pr(X = 4) = \int_4^4 f_X(x) dx = 0.$$

So the probability the government stays in power for exactly 4 years is 0.

- f.

$$\Pr(2 \leq X \leq 4) = F(4) - F(2) = 1 - \frac{2^5}{1024} = 1 - \frac{32}{1024} = 1 - \frac{1}{32} = \frac{31}{32} \approx 0.969$$

Since X has support only on $(0, 4)$, $\Pr(X > 4) = 0$, so

$$\Pr(X < 2 \text{ or } X > 4) = \Pr(X < 2) = F(2) = \frac{32}{1024} = \frac{1}{32} \approx 0.031$$

The two answers sum to

$$\frac{31}{32} + \frac{1}{32} = \frac{32}{32} = 1.$$

This makes sense because $\{2 \leq X \leq 4\}$ and $\{X < 2 \text{ or } X > 4\}$ are complementary events (they partition the full range of X , up to a set of probability zero at the boundary), so their probabilities must add to 1.

Problem 4: Calculating the CDF

a. For $z \geq 0$:

$$F(z) = \int_0^z \lambda e^{-\lambda t} dt = \lambda \left[\frac{e^{-\lambda t}}{-\lambda} \right]_0^z = - \left[e^{-\lambda t} \right]_0^z = - \left(e^{-\lambda z} - e^0 \right) = 1 - e^{-\lambda z}$$

So $F(z) = 1 - e^{-\lambda z}$ for $z \geq 0$, and $F(z) = 0$ for $z < 0$.

b.

$$\Pr(4 \leq Z \leq 9) = F(9) - F(4) = \left(1 - e^{-9\lambda} \right) - \left(1 - e^{-4\lambda} \right) = e^{-4\lambda} - e^{-9\lambda}$$

c. With $\lambda = 2$, solve $F(q) = 0.25$:

$$1 - e^{-2q} = 0.25 \implies e^{-2q} = 0.75 \implies q = \frac{-\ln(0.75)}{2} \approx 0.144$$

d. With $\lambda = 2$:

$$\Pr(Z < 0.75) = F(0.75) = 1 - e^{-2(0.75)} = 1 - e^{-1.5} \approx 1 - 0.223 = 0.777$$

Problem 5: Working with normal random variables

a. Using the standard normal table,

$$\Pr(X \leq 1.28) = \Phi(1.28) = 0.8997$$

For the negative value, use symmetry, $\Phi(-z) = 1 - \Phi(z)$:

$$\Pr(X \leq -0.84) = 1 - \Phi(0.84) = 1 - 0.7995 = 0.2005$$

b. Standardizing with $\mu = 2$, $\sigma = 3$:

$$\Pr(Y \geq 6.5) = 1 - \Pr(Y \leq 6.5) = 1 - \Phi\left(\frac{6.5 - 2}{3}\right) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668$$

c. Since $Y \sim \text{Normal}(2, 9)$ (i.e., $\mu = 2$, $\sigma^2 = 9$, $\sigma = 3$), standardizing gives

$$Z = \frac{Y - 2}{3} \sim \text{Normal}(0, 1)$$

with pdf

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

Problem 6: Relating the Gamma, exponential, and χ^2

- a. Setting $\alpha = 1$ in the Gamma pdf, and using $\Gamma(1) = 1$ and $x^0 = 1$:

$$f(x | 1, \beta) = \frac{\beta^1}{\Gamma(1)} x^{1-1} e^{-\beta x} = \beta e^{-\beta x}$$

Setting $\beta = \lambda$ gives $f(x) = \lambda e^{-\lambda x}$, which is exactly the pdf of an $\text{Exponential}(\lambda)$ random variable. So $\text{Gamma}(1, \lambda) = \text{Exponential}(\lambda)$.

- b. Since $\chi^2(n) = \text{Gamma}(n/2, 1/2)$, for $X \sim \chi^2(6)$,

$$E[X] = n = 6, \quad \text{Var}(X) = 2n = 12$$

Since $X = \sum_{i=1}^6 Z_i^2$ with independent $Z_i \sim \text{Normal}(0, 1)$, and $E[Z_i^2] = \text{Var}(Z_i) = 1$, we have $E[X] = \sum_{i=1}^6 E[Z_i^2] = 6$. For the variance, we use the standard fact that $E[Z_i^4] = 3$ for a standard normal variable, so $\text{Var}(Z_i^2) = E[Z_i^4] - (E[Z_i^2])^2 = 3 - 1 = 2$, giving $\text{Var}(X) = \sum_{i=1}^6 \text{Var}(Z_i^2) = 12$.

- c. Since Y_1 and Y_2 are independent Gamma random variables with the same rate parameter $\beta = 3$, their sum is again Gamma with shape parameters added:

$$Y_1 + Y_2 \sim \text{Gamma}(2 + 5, 3) = \text{Gamma}(7, 3)$$

Using $E[X] = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$ with $\alpha = 7$, $\beta = 3$:

$$E[Y_1 + Y_2] = \frac{7}{3} \approx 2.333, \quad \text{Var}(Y_1 + Y_2) = \frac{7}{3^2} = \frac{7}{9} \approx 0.778$$

Problem 7: Calculating ideal points

- a. Expanding $U(L) = -(L - v)^2 = -(L^2 - 2vL + v^2) = -L^2 + 2vL - v^2$ and applying the expected value rule term by term:

$$E[U(L)] = E[-L^2 + 2vL - v^2] = -E[L^2] + 2vE[L] - v^2$$

Using $E[L^2] = \text{Var}(L) + \mu_l^2$ (where $\mu_l = E[L]$):

$$E[U(L)] = -(\text{Var}(L) + \mu_l^2) + 2v\mu_l - v^2 = -\text{Var}(L) - \mu_l^2 + 2v\mu_l - v^2$$

Grouping the last three terms as a perfect square, $-\mu_l^2 + 2v\mu_l - v^2 = -(\mu_l - v)^2$, so

$$E[U(L)] = -\text{Var}(L) - (\mu_l - v)^2$$

- b. Holding μ_l fixed, the term $(\mu_l - v)^2$ stays fixed, so any increase in $\text{Var}(L)$ directly lowers $E[U(L)]$ (since $\text{Var}(L)$ enters with a negative sign). This means An's expected happiness falls whenever a candidate's future policies become more uncertain, even if the expected policy position does not change. In other words, uncertainty itself is costly.

- c. An's ideal point is $v = 2$. Using $E[U(L)] = -Var(L) - (\mu_L - v)^2$:

$$E[U(L_S)] = -Var(L_S) - (\mu_S - 2)^2 = -8 - (2 - 2)^2 = -8 - 0 = -8$$

$$E[U(L_K)] = -Var(L_K) - (\mu_K - 2)^2 = -3 - (4 - 2)^2 = -3 - 4 = -7$$

Since $-7 > -8$, An votes for **Knope**. Although Schrute's expected policy position coincides exactly with An's ideal point, the high uncertainty surrounding Schrute's policies lowers An's expected utility. Knope's expected policy position is farther from An's ideal point, but it is more predictable. Thus, An prefers Knope despite Schrute having the more favorable expected platform. This illustrates how risk aversion can lead a voter to prefer a candidate with a less desirable expected policy position when that candidate's platform is sufficiently less uncertain.

Problem 8: Moments

- An expected value is the average outcome we would anticipate if we could repeat a random process many times, weighting each possible outcome by how likely it is. It is called the first moment because moments are defined as $E[X^k]$ for $k = 1, 2, 3, \dots$, and the mean is simply this quantity evaluated at $k = 1$, making it the first and most basic member of that family.
- The first moment is the mean itself. We care about it because it gives us a single baseline number for where a distribution is centered, telling us what we should expect to observe on average if our assumptions about the process are correct.
- The second moment is used to build the variance, which tells us how spread out or concentrated the values of a random variable are around the mean. While the first moment tells us the center, the second tells us how much individual outcomes typically deviate from that center.
- The third moment describes skew, or the asymmetry of a distribution (whether one tail is longer than the other), while the fourth moment describes kurtosis, or how heavy the tails are (how likely extreme values are relative to a normal distribution). Together they describe the shape of a distribution beyond just its center and spread.

Problem 9: Functions

- Uniform:** a flat, rectangular shape where every value between a and b is equally likely; $E[X] = \frac{a+b}{2}$.
- Exponential:** peaks at 0 and decays steadily to the right (a long right tail); $E[X] = \frac{1}{\lambda}$.
- Gamma:** takes only positive values and is right-skewed, with a shape that ranges from exponential-like to hump-shaped depending on α ; $E[X] = \frac{\alpha}{\beta}$.
- Normal:** a symmetric bell-shaped curve centered at its mean; $E[X] = \mu$.
- χ^2 : takes only positive values, is right-skewed, and becomes more symmetric (more normal-looking) as the degrees of freedom n grow; $E[X] = n$.

- f. **Student's t:** symmetric and bell-shaped like the normal, but with heavier tails (especially for small n), converging to the normal distribution as $n \rightarrow \infty$; $E[X] = 0$ for $n > 1$.